Unit 6, Lesson 1: Proportional Relationships

Benchmark

MA.8.AR.3.1 Determine if a linear relationship is also a proportional relationship.

Learning Target

- ☐ I can describe attributes of a proportional relationship between two variables represented by tables and equations.

6.1.1 Warm-Up: Day at Work – Bridge Builder

Ken Brown said, "You are hammered all through school to think, and the thinking part is a big part of your education where you are taught how to think and solve problems."

1. Why is learning how to think and solve problems important for any job you do in the future?

6.1.2 Exploration: What is a Proportional Relationship?

The table shows how many cans of disinfectant spray a hospital receives depending on the number of cases ordered.

Cases Ordered	Number of Cans
1	12
3	36
5	60
10	120

1. How many cans are in each separate case when 3 cases are ordered?

2. How many cans are in each case when 5 cases are ordered?

3. How many cans are in each case when 10 cases are ordered?

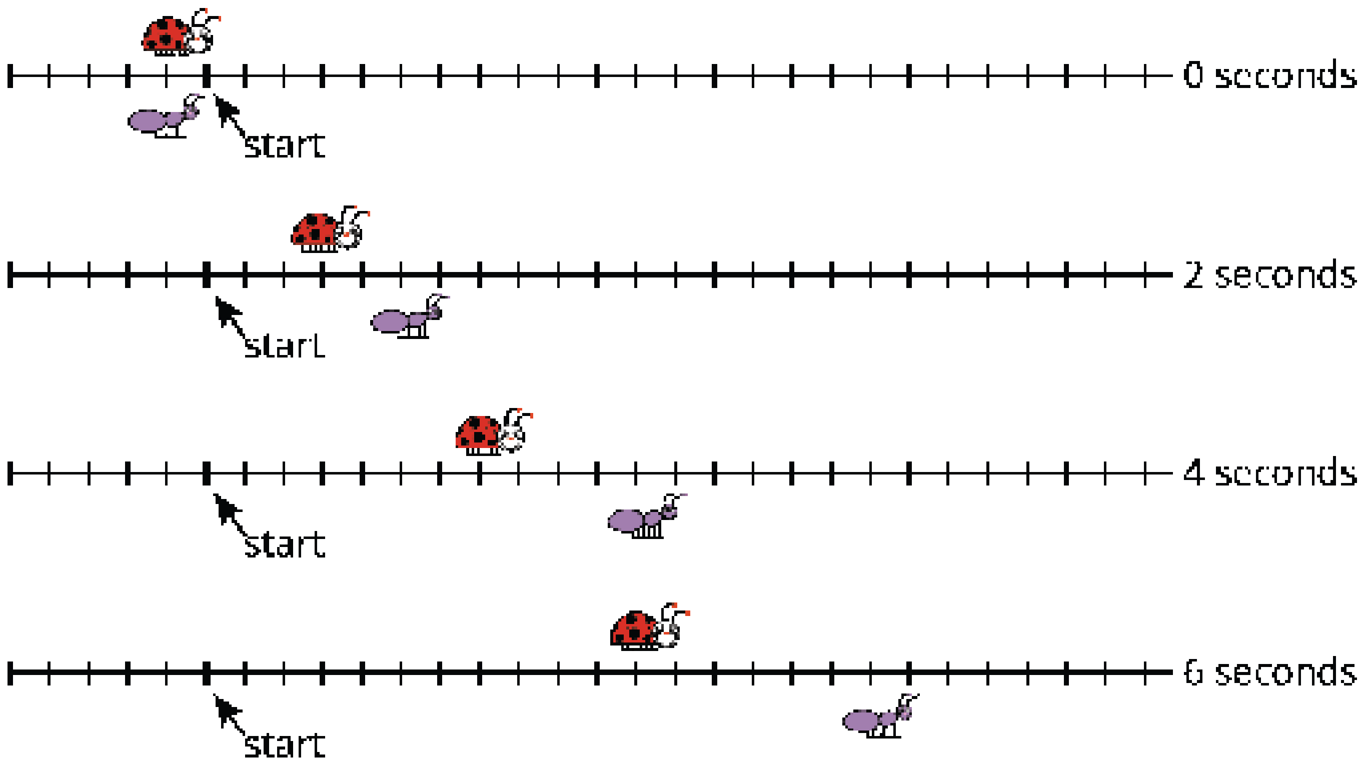
4. Explain how to determine the number of cans of disinfectant spray that would be in each separate case when ordering any number of cases.

5. This situation represents a proportional relationship. Discuss with your partner why the situation is a proportional relationship and summarize your conclusions.

6.1.3 Guided Instruction: Understanding Proportional Relationships

A ***proportional relationship*** is a collection of pairs of numbers that are in equivalent ratios.

A ladybug and an ant move at constant speeds. The diagram with tick marks shows their positions at different times, in seconds (sec). Each tick mark represents 1 centimeter (cm).



1. Complete the table of values for the ladybug and the ant.

Ladybug	
Time (sec)	**Distance (cm)**
0	0
2	
4	
6	

Ant	
Time (sec)	**Distance (cm)**
0	0
2	
4	
6	

One way to determine if a relationship is proportional is to identify if ________________________ ratios exist.

When writing ratios for situations with two variables, each ratio should compare the dependent variable to the independent variable and can be written in fractional form as $\frac{\text{dependent variable}}{\text{independent variable}}$.

The ***independent variable*** is the quantity in the relationship whose values do not depend on changes in other variables. It is typically represented by the variable x.

The ***dependent variable*** is the quantity in the relationship whose values depend on changes in the independent variable. It is typically represented by the variable y.

2. Time and distance are the two variables in this situation. Complete the ratio identifying the dependent and independent variables.

$$\frac{\text{dependent}}{\text{independent}} = \underline{\qquad\qquad}$$

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3. Complete the set of ratios for each insect and determine if the ratios within each set are equivalent.

Ladybug	Ant
$\frac{\quad}{2} =$	$\frac{\quad}{2} =$
$\frac{\quad}{4} =$	$\frac{\quad}{4} =$
$\frac{\quad}{6} =$	$\frac{\quad}{6} =$
Are the ratios equivalent?	Are the ratios equivalent?

4. Explain how you know if the tables each represent a proportional relationship.

5. Discuss with your partner why the ratio, when time is equal to zero, is not determined for each situation and summarize your conclusions.

The ***constant of proportionality***, k, is the constant value of the ratio of two proportional quantities.

6. The equation $y = kx$ can be used to represent any proportional relationship.

   a. Use the word bank to label each part of the equation.

independent variable	dependent variable	constant of proportionality

$$y = k \cdot x$$

   b. What is the constant of proportionality for the ladybug?

   c. Complete the table to explain what the constant of proportionality, k, and the x- and y-variables represent in the context of the ladybug's movement.

Value	Meaning in Context
k	
x	
y	

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d. Write the equation for the ladybug's movement.

e. What is the constant of proportionality for the ant?

f. Complete the table to explain what the constant of proportionality, k, and the x- and y-variables represent in the context of the ant's movement.

Value	Meaning in Context
k	
x	
y	

g. Write the equation for the ant's movement.

h. Discuss with your partner if it is possible to write an equation for a proportional relationship if only the constant of proportionality is given. Summarize your discussion.

i. Use the equation for the ladybug's movement to determine the value of y when $x = 0$.

j. Use the equation for the ant's movement to determine the value of y when $x = 0$.

7. Use the equation $y = k \cdot x$ to explain why the value of y will be 0 when $x = 0$ no matter the value of the constant of proportionality, k.

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6.1.4 Wrap-Up: Cool Down

1. A textbook stated, "We say that y is directly proportional to x if $y = kx$ for some constant k."

   For homework, students were asked to restate the definition in their own words and give an example. Below is the response from one student.

Jessica	When two quantities are proportional, it means that as one quantity increases the other will also increase at the same ratio for all values. An example is the circumference of a circle and its diameter. The ratio of the values always equals π.

   Write your own definition and example, including how to recognize a proportional relationship from a table and an equation.

Name	Definition & Example